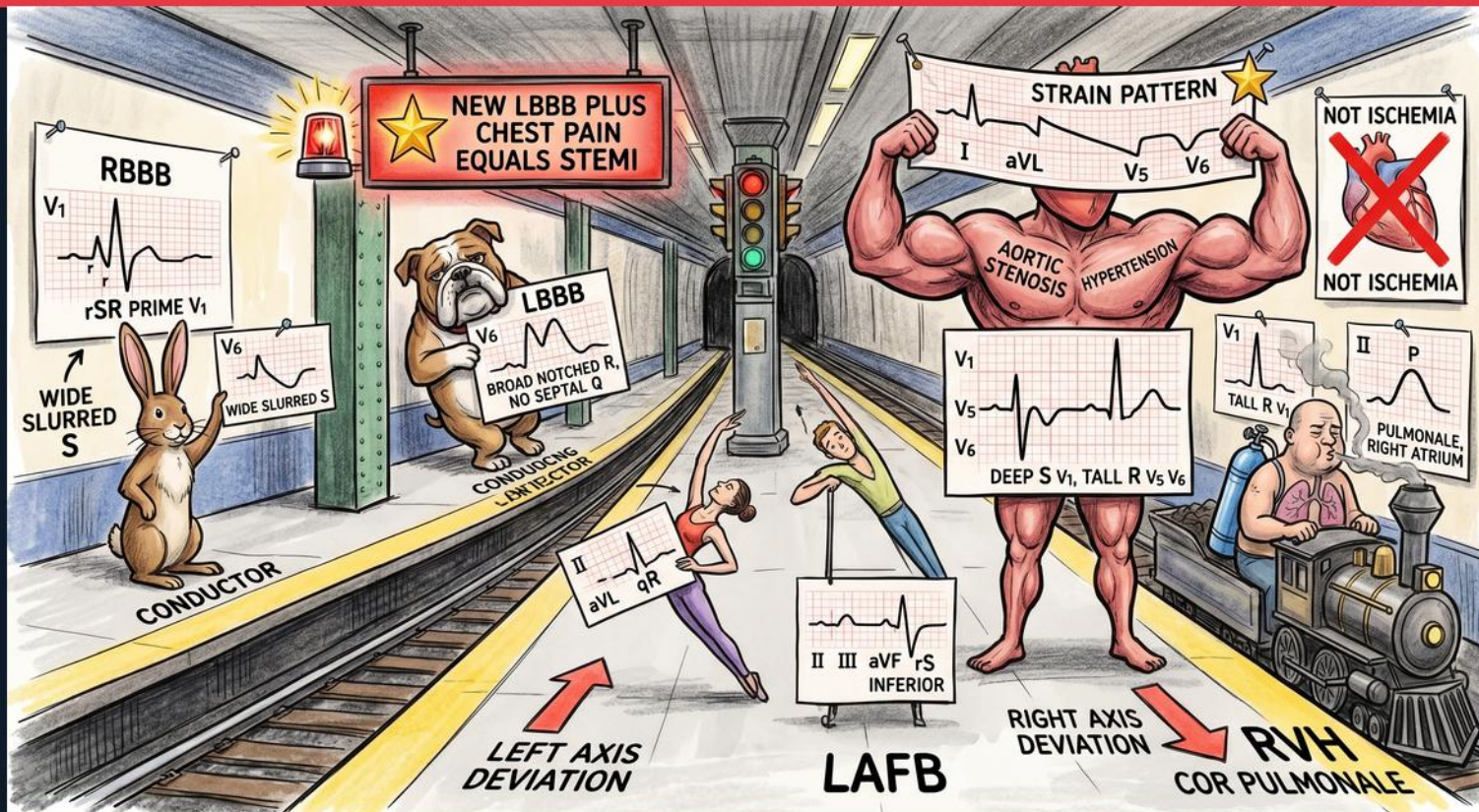
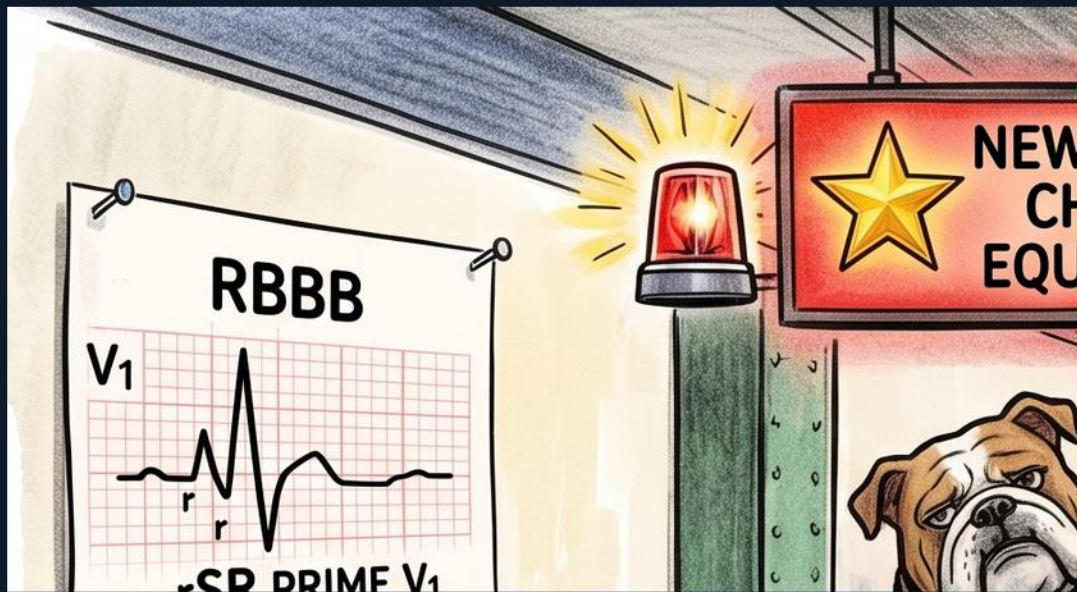


Cardiac Exam — ECG: Conduction & Chambers





RBBB tracing (V1)

WHAT YOU SEE

ECG strip labeled RBBB showing a small r, then S, then a second taller R' in lead V1.

WHAT IT ENCODES

RBBB produces a WIDE QRS (≥ 120 ms) with an rSR' pattern in V1 — the classic 'rabbit ears.' The right ventricle depolarizes late because the right bundle is blocked, so the terminal forces point rightward and anteriorly toward V1. Look for the second R wave (R') in the right precordial leads to call it.

THE BOARD TRAP

Don't read the rSR' as just artifact or a normal variant — a wide QRS with rabbit ears in V1 is a true conduction block.



Rabbit (rSR' rabbit-ears)

WHAT YOU SEE

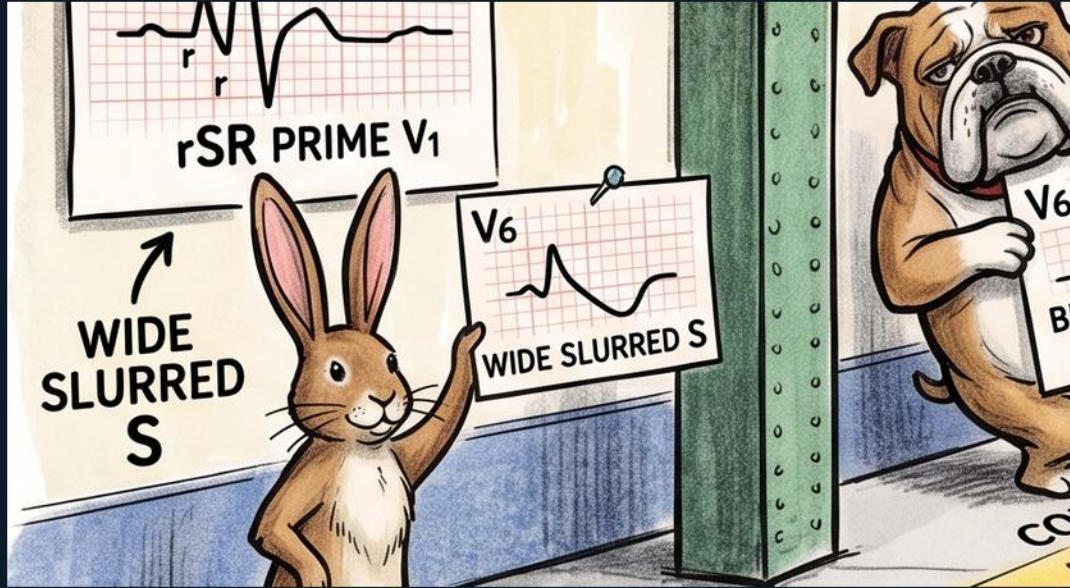
A cartoon rabbit sitting on the platform beneath the RBBB strip, its two upright ears echoing the rSR' shape.

WHAT IT ENCODES

The rabbit anchors the rSR' 'rabbit-ear' morphology of RBBB in V1. Two upright deflections (r and R') separated by the S wave are the visual hallmark. RBBB alone does not block the main left-sided forces, so it rarely hides an MI the way LBBB does.

THE BOARD TRAP

Rabbit ears belong in V1 (right precordial). Expecting them in the lateral leads is wrong — there you instead see a wide, slurred S.



Wide slurred S (V6)

WHAT YOU SEE

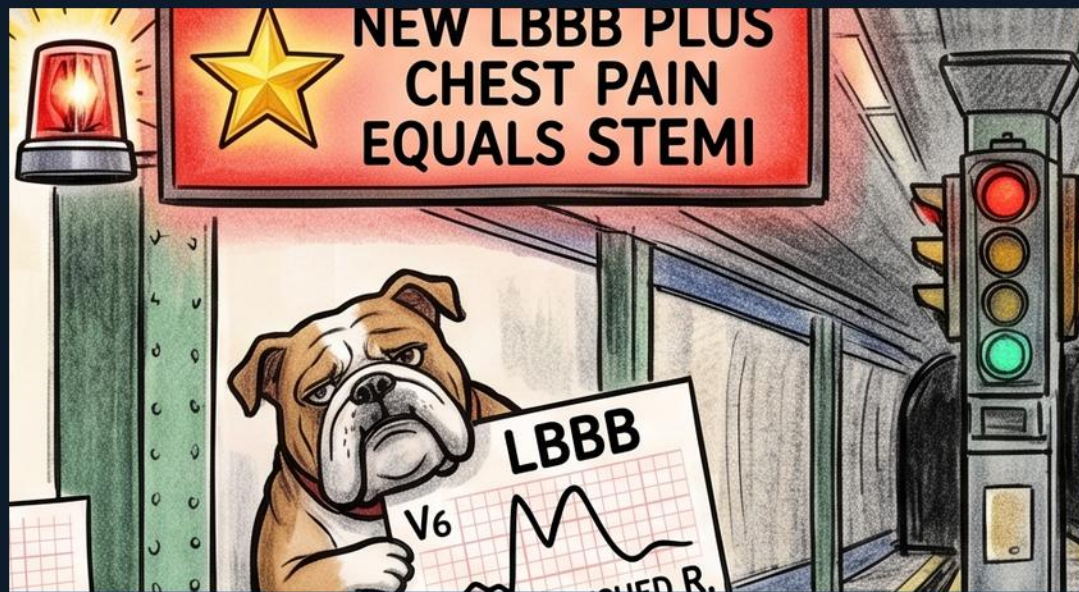
Text 'WIDE SLURRED S' at the far left with a deep, dragged-out S deflection in lead V6.

WHAT IT ENCODES

In RBBB the late right-ventricular forces travel away from the left-lateral leads, so V6 (and lead I) show a WIDE, slurred S wave. This is the lateral-lead counterpart to the V1 rabbit ears. Together — rSR' in V1 plus broad S in V6 — they confirm right bundle branch block.

THE BOARD TRAP

A slurred S in V6 is RBBB, not LBBB — in LBBB V6 shows a broad notched R wave instead, with no S.



LBBB tracing (V6)

WHAT YOU SEE

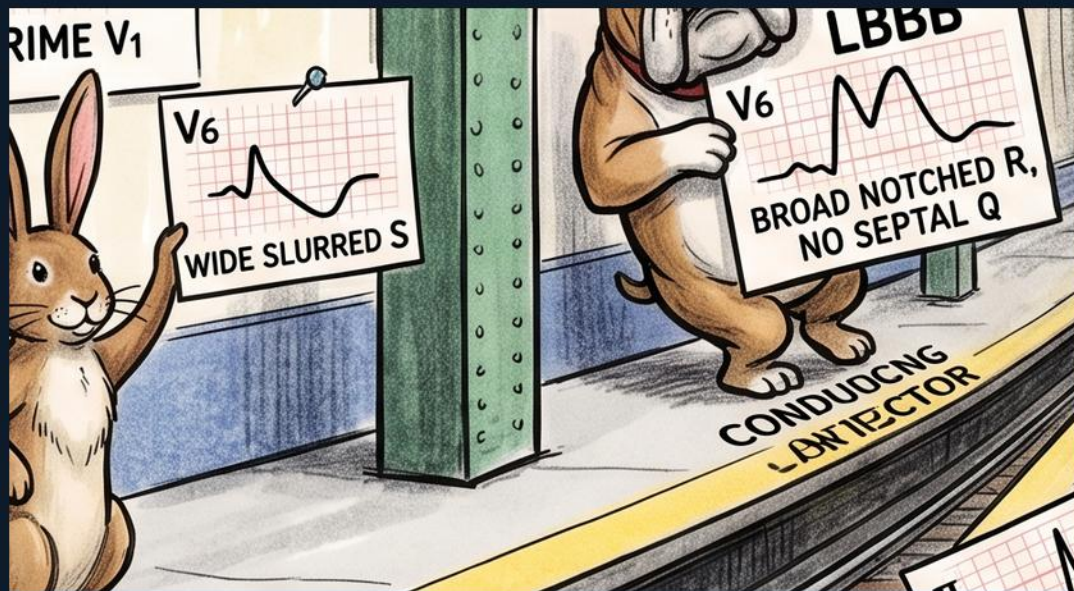
ECG strip labeled LBBB with a broad, notched monophasic R wave in lead V6.

WHAT IT ENCODES

LBBB gives a WIDE QRS with a broad, notched (or slurred) R wave in the lateral leads V6, I and aVL, and characteristically NO septal q wave. The septum normally depolarizes left-to-right, producing that little q; LBBB reverses septal activation, so the q disappears. NEW LBBB with chest pain is treated as a STEMI equivalent.

THE BOARD TRAP

Absence of the normal septal q in I/V6 is expected in LBBB — don't mistake it for an old infarct erasing the q wave.



Bulldog conductor (LBBB)

WHAT YOU SEE

A stout bulldog in a conductor's cap standing on the platform labeled 'CONDUCTOR / LBB BLOCK.'

WHAT IT ENCODES

The bulldog conductor marks LEFT bundle branch block — the blocked 'conductor' of the left-sided tracks. Because LBBB scrambles ventricular activation, you cannot read ST segments for ischemia normally, which is why new LBBB plus symptoms is managed as a STEMI. Sgarbossa criteria are used to find MI within an LBBB.

THE BOARD TRAP

Don't dismiss a stable-looking LBBB tracing in a patient with chest pain — new LBBB + chest pain = activate the cath lab.



NEW LBBB + CHEST PAIN = STEMI banner

WHAT YOU SEE

A red star-burst sign at the top of the station reading 'NEW LBBB PLUS CHEST PAIN EQUALS STEMI,' with a flashing blue light and a red traffic signal nearby.

WHAT IT ENCODES

This is the urgent rule: a NEW (or presumed-new) LBBB in a patient with ischemic chest pain is a STEMI equivalent and warrants emergent reperfusion. The conduction block masks the ST elevation, so you treat on clinical grounds. The red light and alarm flag the time-critical, do-not-wait nature of the call.

THE BOARD TRAP

Waiting for 'classic' ST elevation before acting is the trap — in new LBBB the elevation is hidden, so delay costs myocardium.



LAFB strip — qR in aVL

WHAT YOU SEE

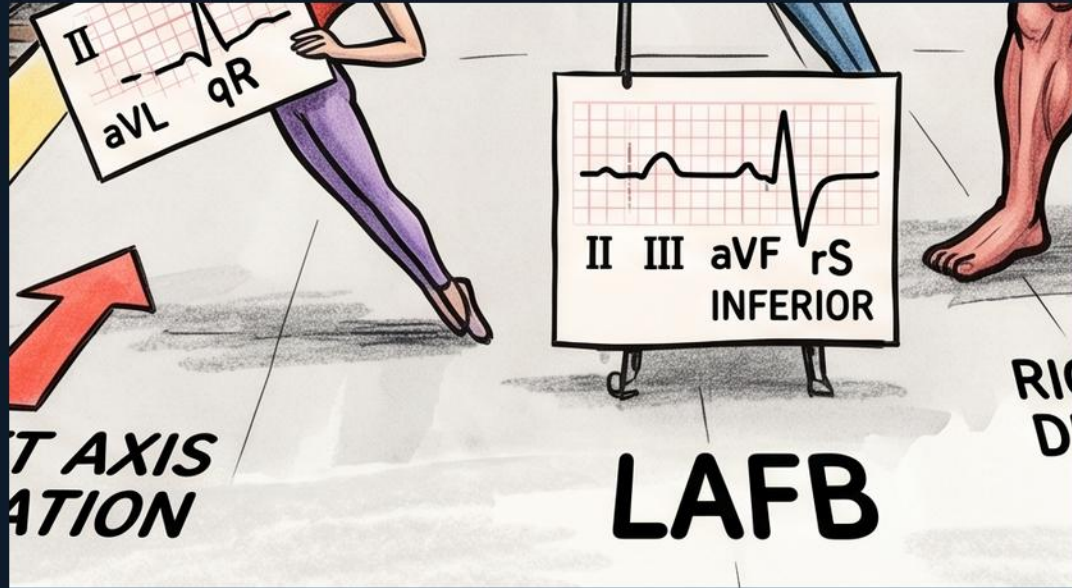
Small ECG card labeled 'II aVL qR' showing a tiny q then tall R in aVL.

WHAT IT ENCODES

Left anterior fascicular block (LAFB) shows a qR complex in lead aVL — a small q followed by an R. Activation reaches the high-lateral wall late and superiorly, giving aVL its qR shape. LAFB is a common, often benign cause of marked left axis deviation.

THE BOARD TRAP

A qR in aVL here reflects fascicular block, not lateral infarction — don't over-call an old MI from the small q.



LAFB strip — rS in II/III/aVF

WHAT YOU SEE

ECG card labeled 'II III aVF rS INFERIOR' showing small r then deep S in the inferior leads.

WHAT IT ENCODES

In LAFB the inferior leads (II, III, aVF) show an rS pattern because the depolarization wavefront moves up and to the left, away from those inferior leads. This rS-inferior / qR-aVL combination is what produces the LEFT axis deviation of LAFB (axis more negative than -45 degrees).

THE BOARD TRAP

rS in the inferior leads is the LAFB signature — don't read the deep inferior S waves as evidence of inferior ischemia.



Yoga dancers — LEFT axis deviation

WHAT YOU SEE

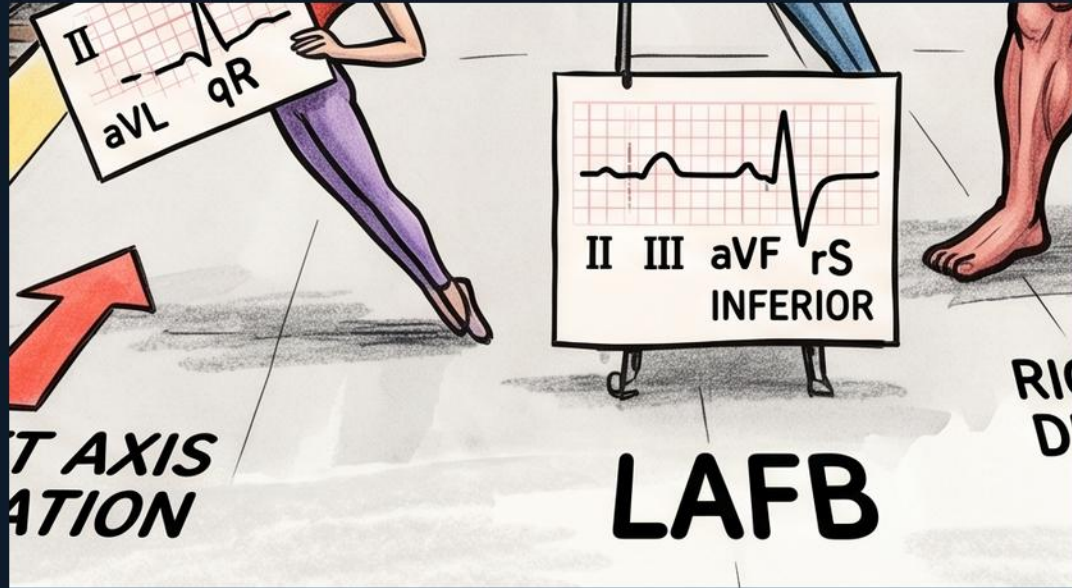
Two flexible dancers bending up and to the left, with a red arrow labeled 'LEFT AXIS DEVIATION' pointing up-left.

WHAT IT ENCODES

The dancers leaning up-and-left depict LEFT axis deviation, the defining feature of LAFB. Axis is leftward (beyond -45°) because the anterior fascicle is blocked and the wavefront swings superiorly. Lead I is upright while II, III and aVF are negative.

THE BOARD TRAP

Left axis deviation has many causes; pairing it with qR-aVL plus rS-inferior is what specifically points to LAFB rather than inferior MI or LVH.



LAFB sign

WHAT YOU SEE

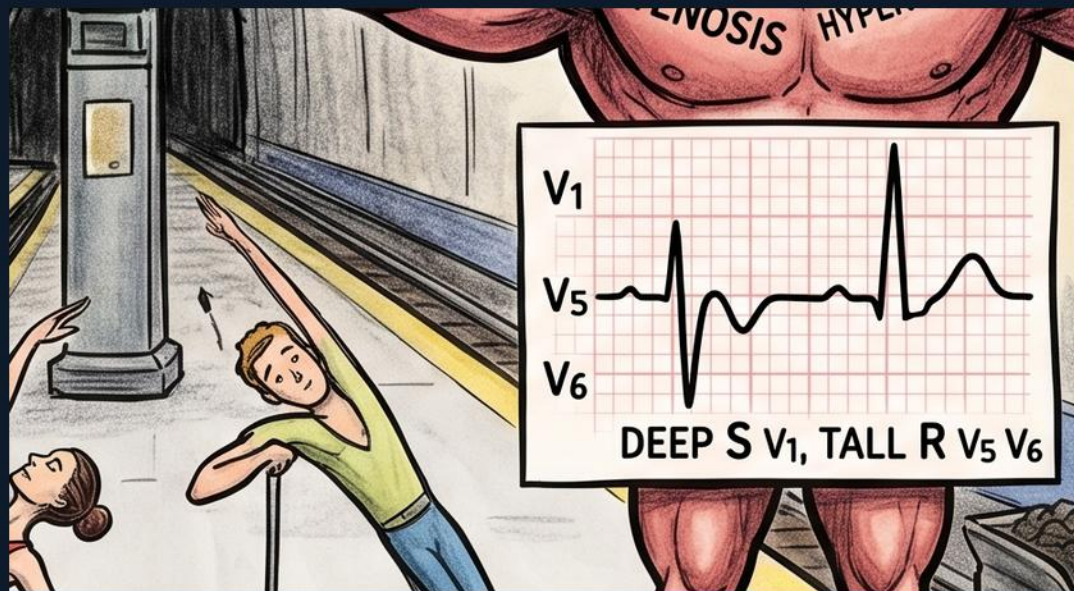
Large platform sign reading 'LAFB' beneath the dancers and inferior-lead card.

WHAT IT ENCODES

LAFB = Left Anterior Fascicular Block. Diagnostic triad: left axis deviation, qR in aVL, and rS in II/III/aVF, with normal or only slightly widened QRS. It localizes the lesion to the anterior division of the left bundle, not the whole left bundle.

THE BOARD TRAP

LAFB widens the QRS minimally; if the QRS is clearly wide with lateral notching, think full LBBB instead of a fascicular block.



Bodybuilder — LVH (deep S V1, tall R V5-6)

WHAT YOU SEE

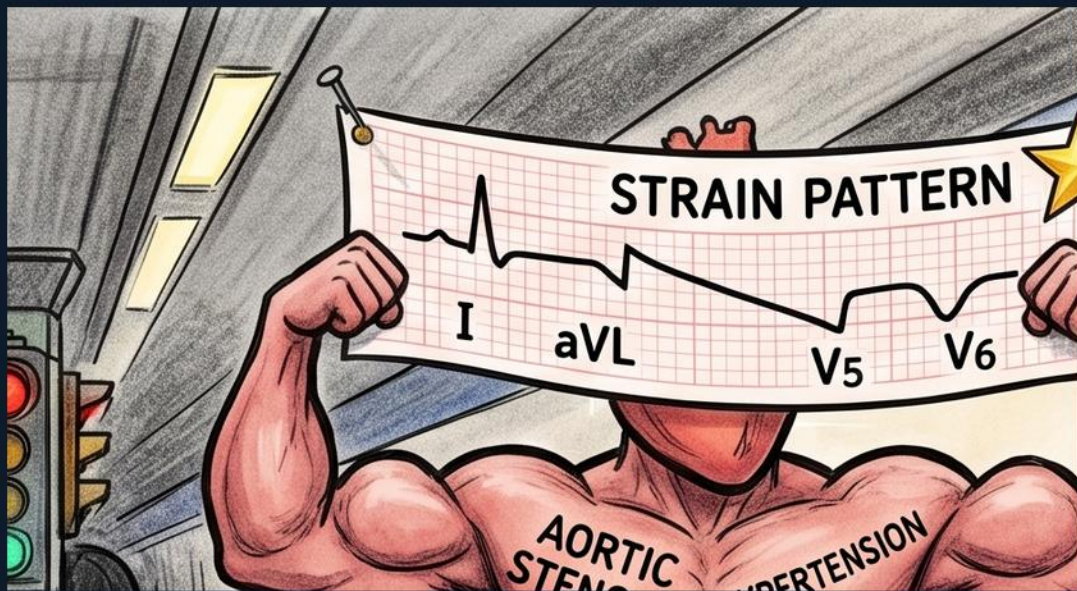
A massive muscular bodybuilder labeled with strips 'V1 ... DEEP S V1, TALL R V5 V6' across his chest, flexing huge biceps.

WHAT IT ENCODES

The hypertrophied muscle man is LEFT VENTRICULAR HYPERTROPHY: high QRS voltage — a DEEP S in V1 plus a TALL R in V5-V6 (Sokolow-Lyon and Cornell criteria). The thick left ventricle generates oversized voltages. Causes shown are aortic stenosis and hypertension.

THE BOARD TRAP

Tall voltages alone are just LVH by voltage; don't automatically add 'strain' unless the ST-T changes are present.



Strain banner (I, aVL, V5, V6)

WHAT YOU SEE

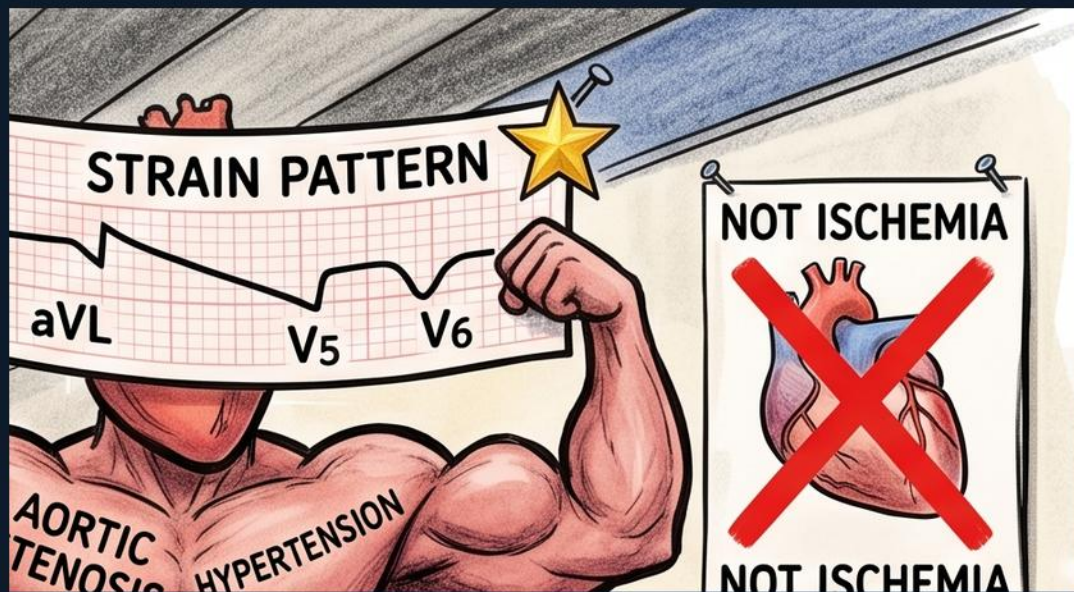
A banner across the top labeled 'STRAIN PATTERN' over strips marked I, aVL, V5, V6 showing downsloping ST depression and asymmetric T inversion, with a gold star.

WHAT IT ENCODES

LVH WITH STRAIN adds downsloping ST depression and asymmetric T-wave inversion in the lateral leads I, aVL, V5 and V6. It reflects the pressure-overloaded thick ventricle (classically aortic stenosis or hypertension), not acute coronary occlusion. Recognizing strain prevents mislabeling these repolarization changes as ischemia.

THE BOARD TRAP

LVH-strain ST/T changes are repeatedly mistaken for ISCHEMIA — the morphology is strain (asymmetric, lateral, with high voltage), so don't call it an NSTEMI.



NOT ISCHEMIA crossed-out signs

WHAT YOU SEE

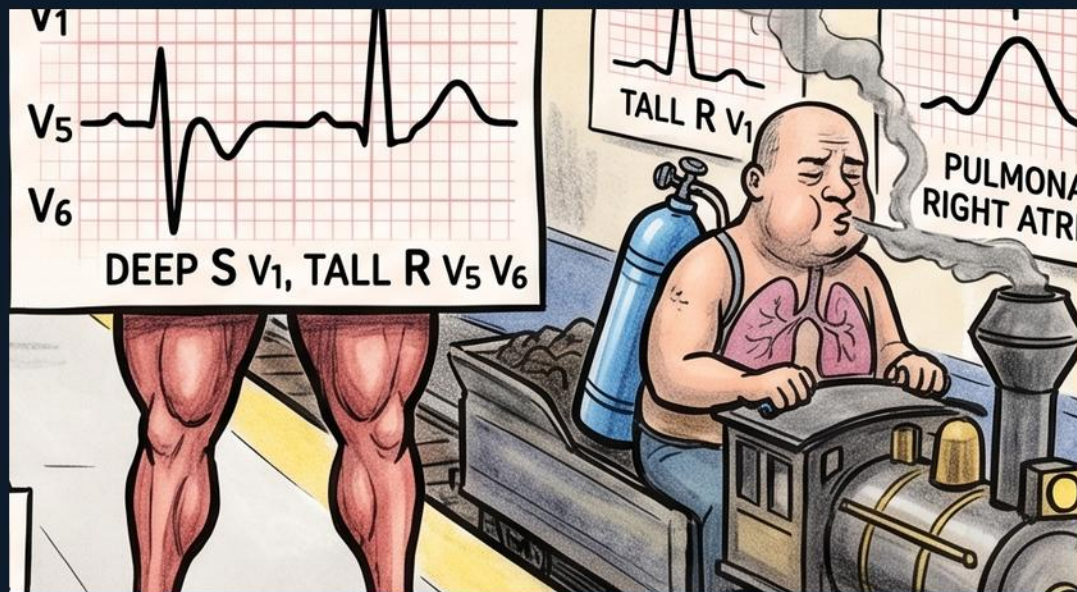
A red X over an ECG with text 'NOT ISCHEMIA' repeated twice on the right wall.

WHAT IT ENCODES

These crossed-out labels hammer the key teaching point: the lateral ST depression and T inversion of LVH with strain are NOT ischemia. They are a secondary repolarization effect of the hypertrophied myocardium. Treat the underlying load (AS, HTN), not as an acute coronary syndrome.

THE BOARD TRAP

Stress-testing or cathing a patient purely for LVH-strain ST-T changes is the trap — strain is a chronic, expected pattern.



Fat man on train — RVH / cor pulmonale

WHAT YOU SEE

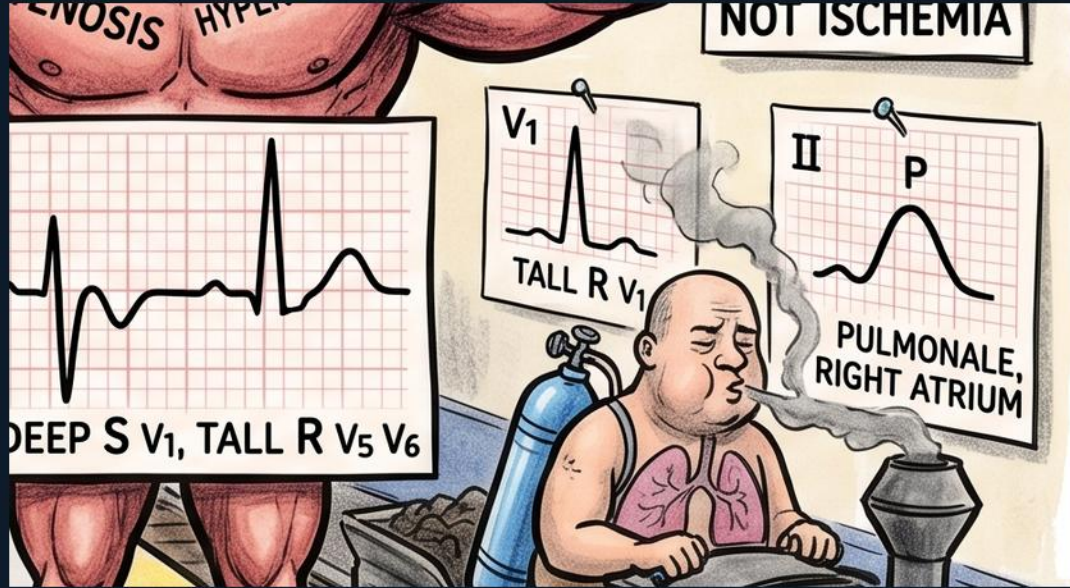
A heavyset man riding a steam train with a red 'RIGHT AXIS DEVIATION' arrow pointing to text 'RVH COR PULMONALE.'

WHAT IT ENCODES

RIGHT ventricular hypertrophy / cor pulmonale shows RIGHT axis deviation and a dominant tall R wave in V1 (the reverse of normal). The pressure-loaded right ventricle (e.g., from chronic lung disease, the smoking steam train) drags the axis rightward. Often accompanied by RV strain T inversions in V1-V3.

THE BOARD TRAP

A tall R in V1 can be RVH, posterior MI, RBBB or WPW — pair it with right axis deviation and P pulmonale to land on RVH.



P pulmonale (tall peaked P in II)

WHAT YOU SEE

Small ECG card labeled 'II P PULMONALE, RIGHT ATRIUM' with a tall, peaked P wave in lead II.

WHAT IT ENCODES

P pulmonale is a tall (>2.5 mm), peaked P wave in lead II signaling RIGHT ATRIAL ENLARGEMENT, the atrial companion of RVH/cor pulmonale. The enlarged right atrium depolarizes first and big, peaking the early part of the P wave. It supports a right-sided pressure-overload picture.

THE BOARD TRAP

Don't confuse P pulmonale (tall, peaked, lead II = right atrium) with P mitrale (wide, notched, M-shaped = left atrium).

Summary — every symbol

1. RBBB tracing (V1)

RBBB produces a WIDE QRS (≥ 120 ms) with an rSR' pattern in V1 — the classic 'rabbit ears.' The right ventricle depolarizes late because the right bundle is blocked, so the terminal forces point rightward and anteriorly toward V1

3. Wide slurred S (V6)

In RBBB the late right-ventricular forces travel away from the left-lateral leads, so V6 (and lead I) show a WIDE, slurred S wave

5. Bulldog conductor (LBBB)

The bulldog conductor marks LEFT bundle branch block — the blocked 'conductor' of the left-sided tracks

7. LAFB strip — qR in aVL

Left anterior fascicular block (LAFB) shows a qR complex in lead aVL — a small q followed by an R

9. Yoga dancers — LEFT axis deviation

The dancers leaning up-and-left depict LEFT axis deviation, the defining feature of LAFB

11. Bodybuilder — LVH (deep S V1, tall R V5-6)

The hypertrophied muscle man is LEFT VENTRICULAR HYPERTROPHY: high QRS voltage — a DEEP S in V1 plus a TALL R in V5-V6 (Sokolow-Lyon and Cornell criteria)

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The rabbit anchors the rSR' 'rabbit-ear' morphology of RBBB in V1

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6. NEW LBBB + CHEST PAIN = STEMI banner

This is the urgent rule: a NEW (or presumed-new) LBBB in a patient with ischemic chest pain is a STEMI equivalent and warrants emergent reperfusion

8. LAFB strip — rS in II/III/aVF

In LAFB the inferior leads (II, III, aVF) show an rS pattern because the depolarization wavefront moves up and to the left, away from those inferior leads

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